

SAI VIJAY KUMAR SURINEELA

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EDUCATION

Masters in Computer Science, May 2026 Wright State University, USA	GPA: 3.625/4
Bachelors of technology in Mechanical Engineering, May 2021 Mahindra University, India	GPA: 8.3/10

TECHNICAL SKILLS

Machine Learning & AI: Regression | Classification | NLP | Transformers | CNN | VAE | DDPM | GAN | Clustering | Generative AI | LoRA | PEFT | RAG | Embeddings | Vector Search
Production ML & MLOps: FastAPI | Docker | MLflow | Model Versioning | CI/CD | GCP | AWS
Data Science & Analytics: Pandas | NumPy | SciPy | Statistical Analysis | A/B Testing | EDA | Data Visualization
Programming: Python | SQL | SOQL | APEX | Java | JavaScript | Bash | Linux
GenAI Stack: LangChain | Hugging Face | Ollama | Vector Databases | Prompt Templates | Embedding Pipelines
Tools: Git | GitHub | Jupyter | VS Code | Linux

EXPERIENCE

Research Assistant Wright State University	Nov 2024 – May 2026 Dayton, OH
- Contributed to 3 machine learning research manuscripts applying data science, statistical analysis, and model evaluation to high-dimensional biomedical and benchmark datasets. - Cleaned, profiled, and prepared datasets with up to 22K+ features, including checks for sample size, feature count, class labels, missing values, class imbalance, and train/test readiness. - Performed EDA across RNA-seq, transcriptomic, biomedical, image, speech, and sensor datasets to analyze feature distributions, class separation, data sparsity, and sample-size limitations. - Designed controlled model-comparison experiments using train/test splits, 5-fold cross-validation, 20 repeated random splits, and baseline benchmarking to evaluate model robustness. - Evaluated predictive performance using accuracy, balanced accuracy, k-NN classification, mean/std across repeated runs, runtime comparison, and error-pattern analysis. - Developed a Block-wise DDPM augmentation workflow for HDLSS biomedical data, using PCA diagnostics, KL divergence, pairwise-distance analysis, and 25 repeated ANN classification trials to validate synthetic data quality across datasets with up to 22K+ features. - Analyzed diffusion-model behavior on biological datasets with limited samples, including experiments on Tabula Muris subsets ranging from 50 to 10K samples.	

Machine Learning Intern Vosyn Inc.	June 2025 – April 2026 Remote
- Implemented the MaskGCT Text-to-Semantic (T2S) training pipeline for a 2-stage TTS architecture, transforming text-token inputs into semantic speech-token representations for downstream acoustic generation. - Built the T2S supervised training workflow by integrating 5 core components: dataset loading, tokenized text inputs, semantic-token targets, model initialization, and training-loop execution. - Prepared and validated clean text-token and semantic-token samples on smaller data batches, resolving dataset-formatting and token-alignment issues before larger-scale T2S training. - Integrated MLflow experiment tracking into the T2S workflow to log 4 training artifacts: parameters, metrics, model checkpoints, and run outputs for reproducible comparison. - Benchmarked Gemma 3 1B optimization techniques for machine translation, measuring GPU memory usage, inference latency, runtime behavior, and output-quality stability. - Implemented BF16 mixed-precision experiments, reducing model memory usage by ~50% after FP32 execution exceeded available GPU memory. - Quantified INT8 and INT4 quantization tradeoffs, reducing model memory by ~34% and ~51%, while identifying hardware-specific inference slowdown on BF16-optimized GPU execution. - Compressed large multilingual translation data from ~98GB to ~26GB using Parquet-based storage, achieving a ~73.5% storage reduction and ~3.8× smaller processing footprint.	

- Extracted and analyzed 52K+ Salesforce CRM records using SOQL, Salesforce Reports, Tableau, and Excel to perform EDA, customer segmentation, SLA trend analysis, and ad hoc business reporting.
- Created 9 derived CRM analytics features, including case frequency, complaint severity, escalation count, customer tenure, repeat-case flag, open SLA status, plan type, region, and average resolution time.
- Standardized 12 CRM reporting fields by converting inconsistent free-text inputs into controlled picklists, required fields, validation rules, and guided Salesforce Flows.
- Built Tableau dashboards on 52K+ Salesforce CRM records to track 8 customer operations KPIs, identifying SLA breach reduction from 18.2% to 14.9%, average resolution time improvement from 39.6 to 34.8 hours, and repeat complaint reduction from 16.4% to 13.7%.
- Applied percentile and IQR-based outlier analysis on case age, resolution time, and SLA breach duration to flag the top 5% long-running cases for escalation-priority review.
- Conducted cohort, pre/post, and ad hoc CRM analysis to compare SLA breaches, repeat complaints, escalation trends, and workflow-impact metrics across customer regions, plan types, complaint categories, and release periods.
- Developed Apex test classes and Jest tests to maintain 85%+ code coverage and support reliable production deployments; coordinated regression validation with QA using Tricentis Tosca.

PROJECTS

Customer Experience Intelligence: NLP-Based Airline Sentiment Analytics

- Built an end-to-end NLP sentiment analytics pipeline on 14,640 airline tweets, performing EDA on tweet-length distributions, missing values, word-frequency patterns, and class imbalance across negative, neutral, and positive sentiment labels.
- Engineered text features using lowercasing, URL removal, punctuation cleaning, stopword filtering, lemmatization, Bag-of-Words, and TF-IDF, generating 10,140-dimensional sparse feature representations for downstream classification.
- Trained and benchmarked 4 sentiment classification models, including Logistic Regression, Naive Bayes, a PyTorch DNN, and an LSTM, achieving up to 78% test accuracy with precision, recall, F1-score, and confusion-matrix evaluation.
- Conducted model error analysis on 654 misclassified tweets to identify class-imbalance effects, neutral-to-negative confusion, sarcasm-related errors, and subtle complaint patterns impacting sentiment prediction quality.

Medical Chatbot using LangChain, Pinecone, and RAG

- Built an end-to-end RAG-based medical question-answering system by parsing healthcare PDFs, cleaning unstructured text, and generating ~5K semantic chunks using 500-token chunk size and 20-token overlap.
- Created dense vector embeddings with Hugging Face Sentence Transformers and indexed them in Pinecone, enabling top-3 semantic retrieval for context-grounded medical QA.
- Integrated LangChain retrieval workflows with OpenAI GPT-4o to generate answers from retrieved medical passages, improving factual grounding and reducing unsupported response generation.
- Evaluated chatbot quality using manually reviewed query-response pairs across 4 criteria: retrieval relevance, answer completeness, groundedness, and hallucination risk; achieved ~85% contextually relevant responses in internal testing.
- Tuned chunking strategy, retrieval depth, prompt templates, and source-context injection to improve answer traceability, semantic match quality, and response consistency across medical-domain queries.
- Productionized the ML application from a notebook prototype into a modular Flask-based RAG pipeline with reusable ingestion, embedding generation, vector search, and answer-generation components.
- Containerized and deployed the application on AWS EC2 using Docker, AWS ECR, IAM, and GitHub Actions CI/CD to automate image builds, registry pushes, and cloud deployment updates.

CERTIFICATIONS

- Salesforce Certified Platform Developer I (PD1)
- Salesforce Certified Platform App Builder